

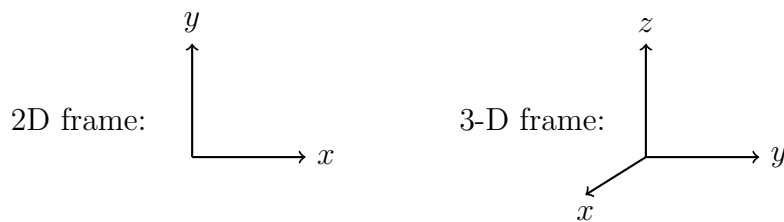
Manipulator Kinematics 101

written by a retarded fuck

Lets start simple. Before we start moving the end effector with 6 different joints in series, let's first try to represent rotation and translation of one joint better.

First, instead of points in a 3-D graph, we use frames. This is because, we want to both know and control both the position and orientation. A point is only defined by its $[x, y, z]$ coordinates and only tells us about position.

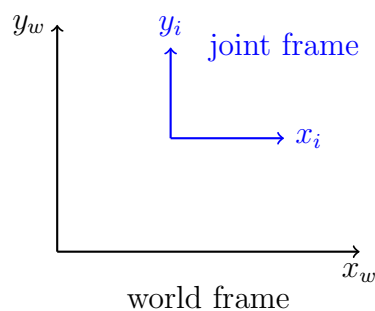
A frame is just a smaller coordinate system.



Okay lets now represent our one joint as a frame. To keep it simple, lets do it in 2-D. We have our joint frame. But we are telling its coordinates with respect to what? We need something, something constant and unmoving to compare its position and orientation with. (Similar to how you use the origin as a reference for plotting points in graph).

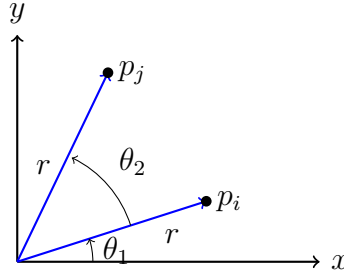
** pose refers to both position & orientation combined.*

So lets have another frame called “the world frame”.



Great! Now we just need an efficient way to represent the “pose” of this joint frame with respect to world frame.

Position/translation is easy – its just the x & y coordinates added to the reference. Rotation/orientation on the other hand – lets take a small example.



Lets take a point p_i in a 2D graph. Let its coordinates be x_i & y_i . Now, I'm gonna rotate it by an angle θ_2 about the origin from its current position. Lets call the new position p_j . How to find p_j using p_i & θ_2 ?

Lets represent p_i & p_j in polar coordinates.

$$p_i = \begin{bmatrix} r \cos \theta_1 \\ r \sin \theta_1 \end{bmatrix} = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \quad p_j = \begin{bmatrix} r \cos(\theta_1 + \theta_2) \\ r \sin(\theta_1 + \theta_2) \end{bmatrix}$$

Expanding p_j ,

$$\begin{aligned} p_j &= \begin{bmatrix} r[\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2] \\ r[\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2] \end{bmatrix} \\ &= \begin{bmatrix} r \cos \theta_1 \cos \theta_2 - r \sin \theta_1 \sin \theta_2 \\ r \sin \theta_1 \cos \theta_2 + r \cos \theta_1 \sin \theta_2 \end{bmatrix} \\ &= \begin{bmatrix} x_i \cos \theta_2 - y_i \sin \theta_2 \\ y_i \cos \theta_2 + x_i \sin \theta_2 \end{bmatrix} = \begin{bmatrix} x_j \\ y_j \end{bmatrix} \end{aligned}$$

Huh, we have now represented the rotation operation with just the initial position and the rotation angle. But wait, we can do one more thing to make it more elegant.

$$\begin{bmatrix} x_j \\ y_j \end{bmatrix}_{2 \times 1} = \begin{bmatrix} x_i \cos \theta_2 - y_i \sin \theta_2 \\ x_i \sin \theta_2 + y_i \cos \theta_2 \end{bmatrix} = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_i \\ y_i \end{bmatrix}_{2 \times 1}$$

Yeah, its just a matrix multiplication. We call the 2×2 matrix "R" $\rightarrow$ The rotation matrix.

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}_{2 \times 2} \quad \text{for 2D}$$

where θ is angle of rotation.

Now I know how my coordinates changed due to rotation. And translation is just adding the displacements in each direction to the current coordinates. So, we can now write a general equation to characterize any rotation/translation.

Let $X_j = \begin{bmatrix} x_j \\ y_j \end{bmatrix}$ be final coordinates. $X_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}$ be initial coordinates. R is rotation matrix. Let P be translation matrix $\begin{bmatrix} P_x \\ P_y \end{bmatrix}$ where P_x, P_y are displacements along the respective axes.

$$\boxed{X_j = RX_i + P} \quad \text{--- (1)} \tag{1}$$

But wait, we can make this also a single matrix multiplication.

$$\begin{bmatrix} x_j \\ y_j \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} P_x \\ P_y \end{bmatrix}$$

If we just add a dummy row,

$$\begin{bmatrix} x_j \\ y_j \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{11} & R_{12} & P_x \\ R_{21} & R_{22} & P_y \\ 0 & 0 & 1 \end{bmatrix}}_{\text{transformation matrix } T} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

We call this new representation of a point $(x_i, y_i, 1)$ as homogenous coordinates.

But wait, why is representing it as a matrix multiplication better? Seems like equation (1) was elegant enough. But matrix representation gives us the ability to stack any number of transformations. We can just keep multiplying transformation matrices and the operations stack!

Let say I first transform my point by T_1 , then again by T_2 and finally by T_3

$$X_j = T_3 T_2 T_1 X_i$$

See? Elegant as fuck. And the best part is the computers process matrix multiplications much faster. So speeded!

For R , lets describe a rotation matrix for rotation about each of the three axes. In 2-D, it was simple cuz we just rotated about z-axis. Now we have 3 different types of rotations possible.

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So final rotation matrix is $R = R_z(\gamma)R_y(\beta)R_x(\alpha)$

where α, β, γ are angles of rotation about x, y, z axes respectively.

**Don't memorize the final R , it changes based on convention.
Just understand the logic**

So then for 3D,

$$\begin{bmatrix} x_j \\ y_j \\ z_j \\ 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & P_x \\ R_{21} & R_{22} & R_{23} & P_y \\ R_{31} & R_{32} & R_{33} & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ z_i \\ 1 \end{bmatrix}$$